Predicting Economic and Population Trends across Counties in the United States

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The purpose of this study was to investigate and model the economic, demographic, and geographic factors that are associated with growth at the county level in the United States using linear regression and a feed-forward neural network. In the context of this study, growth was defined as the percent change in population, employment, number of hiring establishments, and gross domestic product (GDP) for each county in the contiguous United States. This study aims to identify which factors are most predictive of rapid job growth and population growth in the United States. It serves to provide valuable insight to urban planners, policymakers, businesses, and investors seeking to understand the drivers of county-level and metropolitan growth. Identifying these variables will serve to further frame future economic realignment and migratory patterns across the United States. With the increasing complexity of these migration patterns and labor market dynamics, there is a critical need for data-driven analysis that can predict and explain where and why growth is occurring.

To address this challenge, this study developed and compared predictive models to quantify the relationship between a wide range of county-level features and multiple independent growth indicators. The objective was to identify which factors are most predictive of rapid population and job growth, producing interpretable and generalizable models for different stakeholders to guide their economic and policy decisions in different regions in the United States. This analysis provides a framework for targeted, evidence-based decision-making to better align economic planning efforts with actual growth trends observed in United States counties in recent years. By determining which local attributes (such as wage increases, educational attainment, and regional geography) are most strongly associated with various types of county-level growth, stakeholders can prioritize different regions for investment and adjust urban development strategies.

Methods

Data Source and Description

The data for this study was aggregated from multiple government agencies, primarily from the Bureau of Economic Analysis and the Bureau of Labor Statistics. These datasets contained population and economic data for each county in the United States through 2023, including the population, current employment, number of businesses, GDP, average salary, and wage increases each year. Additionally, United States Census data was collected to add data corresponding to the percentage of each county's population that is college-educated and the percentage of each county's population that is unemployed. The dataset was further augmented with average temperature data from the National Centers for Environmental Information. Lastly, the median household income and median annual cost of living for a household size of three in each county was identified from data collected by the independent think-tank the Economic Policy Institute and added to the dataset.

Data Preprocessing

The data was augmented with four features to be the outcome variables that the models would predict: Population Change Percentage, Employment Change Percentage, Establishment Change Percentage, and GDP Growth Percentage. These features were calculated by taking the difference between the yearly totals for each of these variables between 2023 and 2019, and dividing this difference by the total value in 2019. Counties with missing values for any feature were dropped as due to the large differences between counties, values could not be reasonably imputed. Correlations were calculated between each feature to determine any multicollinearity, and one feature in each pair with a correlation of 0.8 or greater were dropped. Categorical variables were one-hot encoded to be used in regression analysis. Outliers were preserved in the dataset, as these large variations were determined to be reflective of real-world patterns and were of interest. The remaining dataset contained 3,044 observations with 64 input features and 4 output features. Of the 3,144 counties in the United States, 3,044 counties from the contiguous United States were left in the cleaned dataset, so 96.8% of the data was retained.

Baseline Analysis: Multi-output Linear Regression

A multi-output linear regression was used as a baseline model to establish a relationship between the input features and the outcome variables. LinearRegressor and MultiOutputRegressor were imported from Scikit-learn and trained on scaled data using StandardScaler. This model was chosen because its coefficients are easily interpretable and can provide a direct quantification of the impact of each input variable on the output variables. Furthermore, many county-level economic and demographic variables were expected to have an additive and linear relationship with growth indicators in each county, like population and economic growth, so a linear regression model allowed for straightforward estimation of the marginal effects of each feature on growth outcomes.

While a standard linear regression model provides a useful baseline for understanding the relationship between county-level predictors and growth outcomes, it is sensitive to overfitting and multicollinearity. A Ridge regression model was therefore generated to extend the baseline by applying L2 regularization, penalizing overfitting and reducing coefficient variance, granting more stable and generalizable predictions. This Ridge regression model was used to reduce the impact of any multicollinearity not caught in the data preprocessing and further maintain the interpretability of the linear model in the presence of many correlated socioeconomic and demographic variables. The alpha coefficient was a hyperparameter associated with the Ridge regression model that controlled the strength of L2 regularization. Alpha values of 0.01, 0.1, 1.0, 10, 100, and 1000 were considered and tested using 5 fold cross validation from the GridSearchCV hyperparameter tuning tool, and the coefficient of determination for each model was compared.

Alternative Analysis: Feed-forward Neural Network

To complement and compare against the linear models, a feed-feed forward neural (FFNN) network was developed using the Sequential API imported from Keras/Tensorflow. This alternative model was used because neural networks are capable of learning complex and nonlinear relationships between input features and target variables that the linear regression model may have failed to capture. Unlike the additive, linear relationship assumed by the linear regression model, the neural network uses layers of weighted transformations and nonlinear activation functions that allow more flexible mappings from input features to growth outcomes.

Different architectures were compared with two dense layers, ReLU activation, and the Adam optimizer, but the learning rate and number of neurons in each layer were varied. To prevent overfitting, a 0.2 validation split and early stopping were used during training with mean absolute error (MAE) loss, which was appropriate for a continuous regression task and robust to outliers preserved in the training data. The model was trained on the same outcome variables as the linear and Ridge regression models to allow direct performance comparison.

While neural networks lack the direct interpretability of linear models, they are able to capture nonlinear interactions among geographic, demographic, and economic predictions. This makes feed-forward neural networks a useful tool for evaluating whether more complex, more powerful models can substantially improve the predicting power over simpler, more interpretable baseline models.

Training and Evaluation Paradigm

Training and testing data were generated using an 80/20 split. The continuous input features were scaled and used to train each model, but the output features were scaled only for the training and testing of the FFNN. The coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) were evaluated and compared for each outcome variable in each model. The R^2 score was used to determine the proportion of variance in each outcome variable each model can explain. It further provides a normalized and easily interpretable metric for comparing how well different models capture the underlying structure in the data. RMSE measures the square root of the average squared differences between predicted and actual values. While sensitive to outliers, it was appropriate to evaluate how each model can handle the vast real-world extremes in growth across different US counties. MAE was used as it is robust to outliers, capturing the average magnitude of prediction errors, allowing us to evaluate how far predictions are, on average, from the true values in the same units as the target variable. Using these three metrics, this analysis was able to balance variance explanation, error magnitude sensitivity, and robust error interpretation, enabling a reliable and multi-faceted assessment of model performance across different growth outcomes for counties in the United States.

Additionally, the coefficients contributing to the Ridge regression model were analyzed to determine which factors were contributing to the models predictions, and thus providing insight into what factors specifically drive growth at the county level. To interpret the predictors contributing to the neural network, SHAP values were analyzed using DeepExplainer and SHAP summary plots. While a linear regression model is very easy to interpret as its coefficients maintain an additive relationship with its predictions, the coefficients of a neural network are complicated weights and biases that are challenging to interpret. SHAP values allow us to analyze specifically how much the values of each input feature influenced the predicted value of each output feature.

Results

The goal of this analysis was to identify key predictors of economic and demographic growth across U.S. counties using multiple regression models. A baseline multi-output linear regression model was used to establish a relationship between input features and growth outcomes, followed by a Ridge regression and a feed-forward neural network to explore the impact of regularization and nonlinear modeling. This section presents the performance metrics and interpretation of each model, comparing their predictive insights across four outcome variables representing county-level growth: population change, employment change, establishment change, and GDP growth.

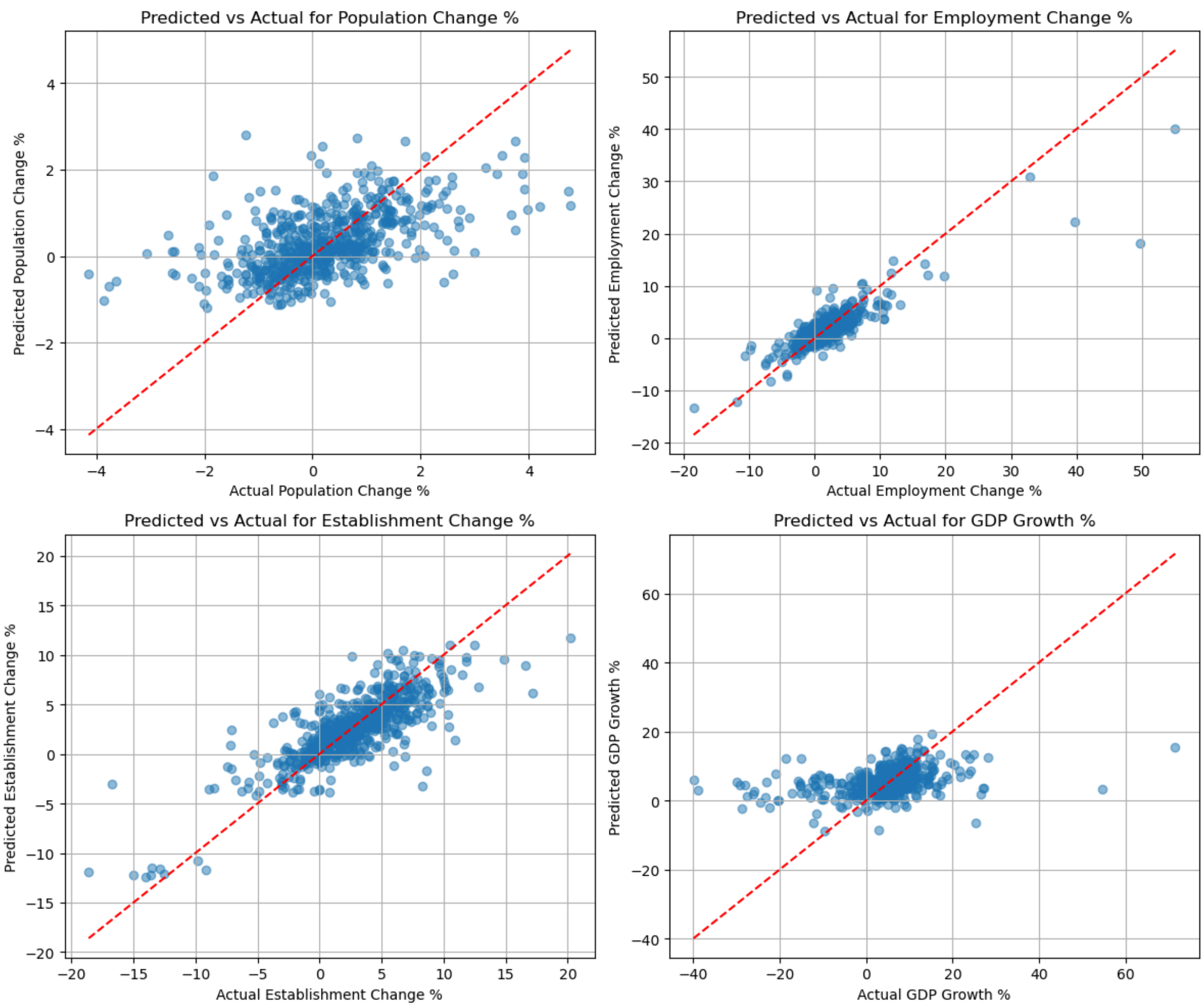

Figure 1: Baseline Multi-Output Linear Regression Model

Table 1: Performance Metrics for Baseline Linear Regression

Outcome Variable	R^2 Score	RMSE	MAE

Table 1: Performance Metrics for Baseline Linear Regression			
Population Change %	0.282164	0.996744	0.723448
Employment Change %	0.738500	2.457727	1.451059
Establishment Change %	0.548143	2.868853	2.137042
GDP Growth %	0.040408	8.848540	5.667106

Figure 1 and Table 1 show the predictions and performance metrics of the baseline linear regression model. As seen in Table 1, the coefficient of determinations for each outcome variable were as follows: 0.282 for population growth, 0.739 for employment growth, 0.548 for establishment growth, and 0.040 for GDP Growth. This indicates limited ability to capture variability in outcomes for population and GDP growth, and moderate ability for variability in outcomes for employment and establishment growth. Ridge regression notably improved these metrics.

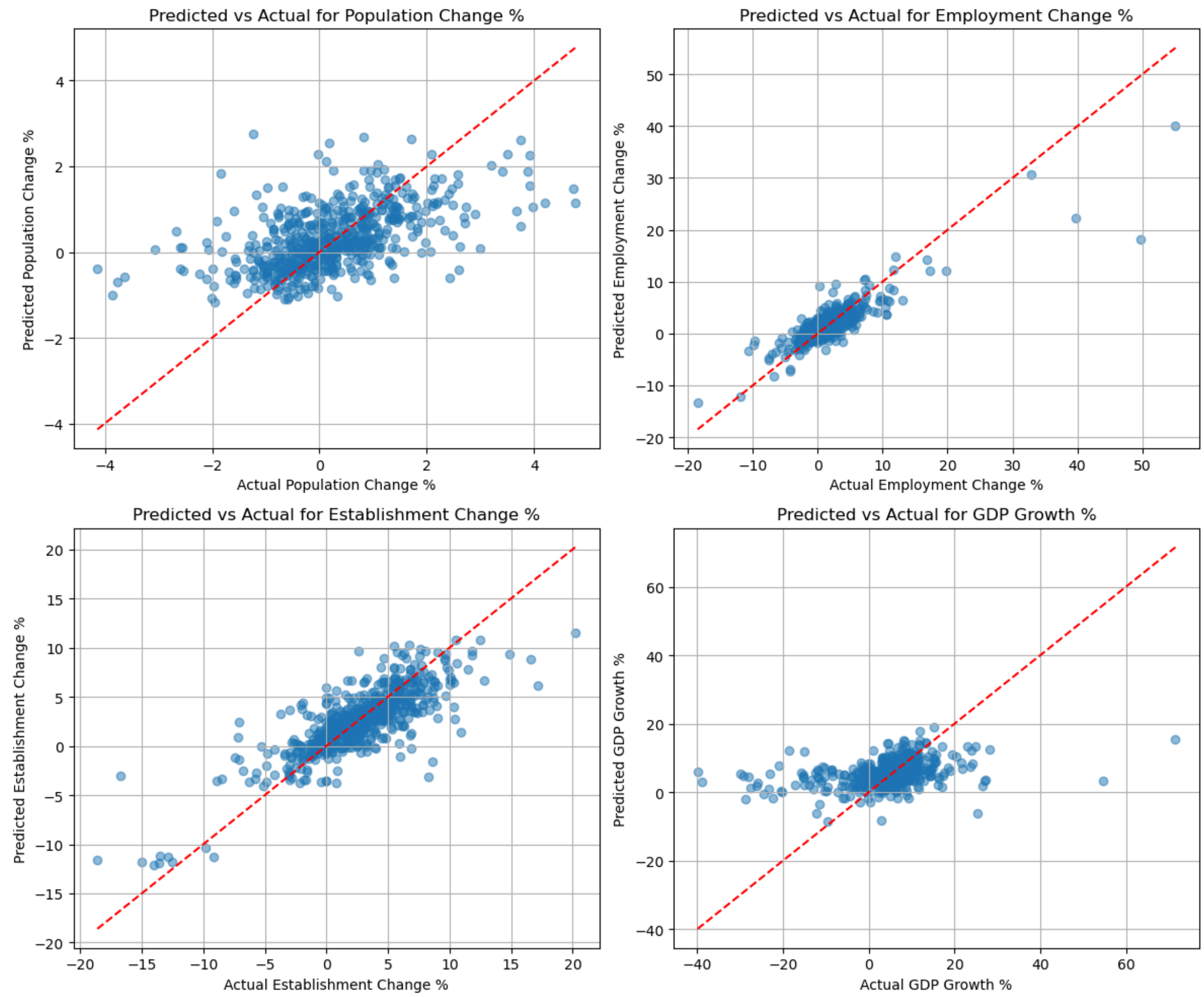

Figure 2: Ridge Regression

Table 2: Performance Metrics for Ridge Regression Model			
Outcome Variable	**R² Score**	**RMSE**	**MAE**

Table 2: Performance Metrics for Ridge Regression Model			
Population Change %	0.284681	0.994995	0.721680
Employment Change %	0.740672	2.447498	1.439627
Establishment Change %	0.647896	2.532462	1.804356
GDP Growth %	0.122403	8.462055	5.075576

Figure 2 and Table 2 show the predictions and performance metrics for the outcome variables for the Ridge regression model. This model captured a similar amount of variability in population growth with a R^2 of 0.285 compared to 0.282 in the linear model, but improved performance for establishment and GDP growth to 0.648 and 0.122 respectively. Additionally, error was reduced for each outcome variable between the linear regression model and the Ridge regression model.

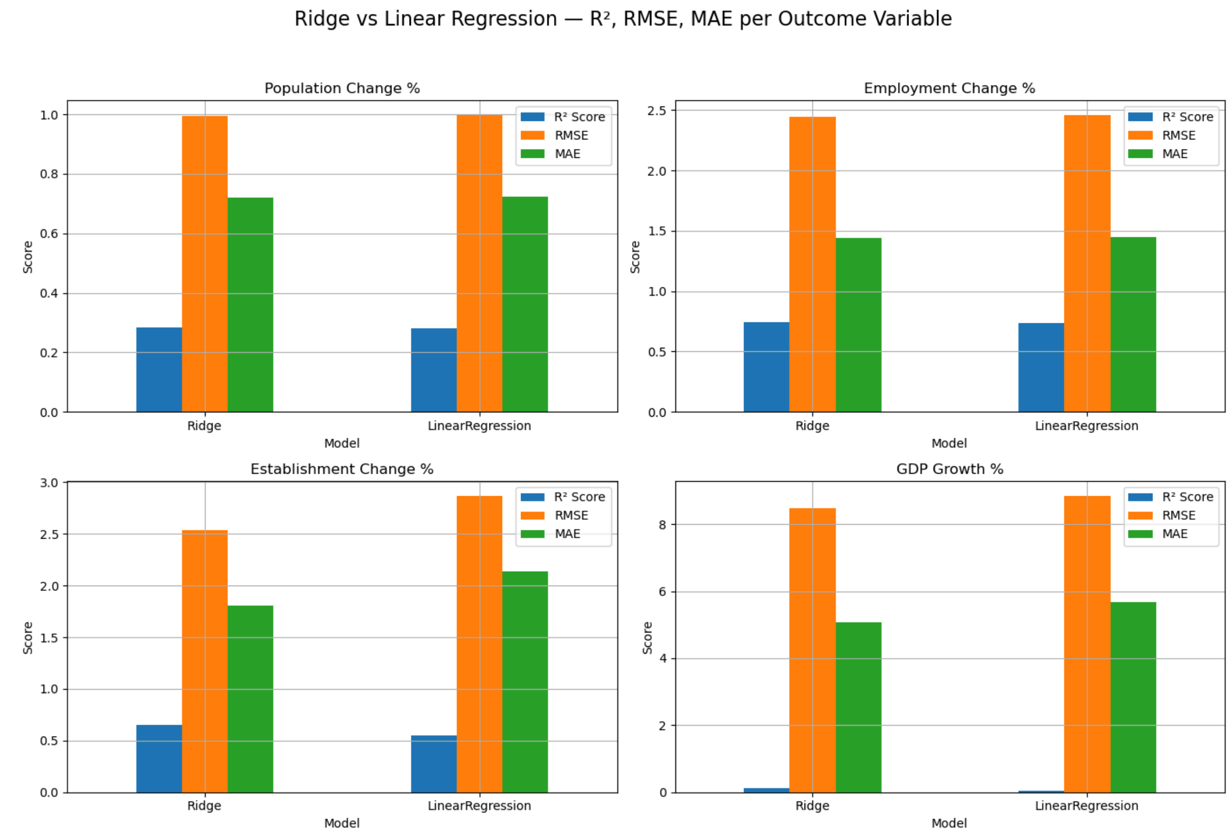

Figure 3: Performance Metrics for Linear Regression and Ridge Regression

Figure 3 highlights the performance differences between the linear regression model and the Ridge regression model. The bar chart shows that both models performed similarly when predicting population growth and employment growth, but the Ridge model overall had higher R^2, lower RMSE, and lower MAE for each outcome variable.

Table 3: Evaluation of Ridge Regression Performance Against Mean Predictions			
Outcome Variable	**Baseline MAE**	**Model MAE**	**MAE Improvement (%)**
Population Change %	0.888069	0.721680	18.736106

Table 3: Evaluation of Ridge Regression Performance Against Mean Predictions			
Employment Change %	2.464667	1.439627	41.589375
Establishment Change %	3.130143	1.804356	42.355471
GDP Growth %	5.478657	5.075576	7.357294

Table 3 shows the results of comparing the mean absolute error between the Ridge regression model and a baseline model that simply predicts the mean to determine the prediction power of the regression model. While only performing slightly better than the mean for GDP growth (7.357% improvement), the regression model's predictions were ~42% more accurate for predicting employment and establishment growth.

Table 4: Most Significant Ridge Regression Coefficients: Population Growth			
All Features		**Continuous Only**	
State_FL	1.527537	**Median Family Income**	0.521985
State_TN	1.040068	**Average Temperature**	0.233248
State_ID	0.965791	**Pct Wage Change**	0.114099
State_SC	0.898716	**Average Annual Pay**	-0.102862
State_LA	-0.883481	**Unemployment**	-0.046289
State_DE	0.814602	**Percent w/ Bachelor's+**	-0.045843
State_NC	0.783386	**Annual Cost**	-0.044173
State_ME	0.705301	**Population**	-0.021230

Table 5: Most Significant Ridge Regression Coefficients: Employment Growth			
All Features		**Continuous Only**	
Pct Wage Change	3.280149	**Pct Wage Change**	3.280149
State_WV	-1.519924	**Percent w/ Bachelor's+**	0.295428
State_VA*	-1.078966	**Annual Cost**	0.263051
State_VA	-1.054405	**Average Annual Pay**	-0.178021
State_LA	-1.024583	**Average Temperature**	0.170856
State_MI	0.859744	**Unemployment**	-0.108003
State_DE	0.855603	**Median Family Income**	-0.037127
State_MT	-0.808149	**Population**	0.011027

Table 6: Most Significant Ridge Regression Coefficients: Establishment Growth			
All Features		**Continuous Only**	
State_WA	-10.187388	**Average Temperature**	0.573019
State_VA	-6.565687	**Percent w/ Bachelor's+**	0.555905
State_WV	6.015718	**Pct Wage Change**	0.415679
State_MT	4.895744	**Average Annual Pay**	-0.216961
State_NE	-4.893117	**Median Family Income**	0.211772
State_MI	4.877325	**Unemployment**	-0.059704
State_VA*	-4.538239	**Annual Cost**	0.039903
State_AR	-4.527849	**Population**	-0.012103

Table 7: Most Significant Ridge Regression Coefficients: GDP Growth			
All Features		**Continuous Only**	
State_ND	-7.590844	**Percent w/ Bachelor's+**	1.997414
State_NE	7.479045	**Average Annual Pay**	-1.317023
State_KS	4.799314	**Pct Wage Change**	1.208412
State_IL	-3.466501	**Unemployment**	0.602948
State_FL	3.299109	**Annual Cost**	0.360443
State_IA	-3.089584	**Median Family Income**	-0.309278
State_OK	-2.774417	**Population**	0.290518
State_VT	-2.524132	**Average Temperature**	-0.124137

Tables 4, 5, 6, and 7 show the most significant coefficients in the Ridge regression model for each growth indicator, divided between all features and only continuous features. The most predictive features for each variable are the states each county is located in typically, but limiting the coefficients to only the continuous variables the model was trained on reveals what other factors are predictors for each outcome.

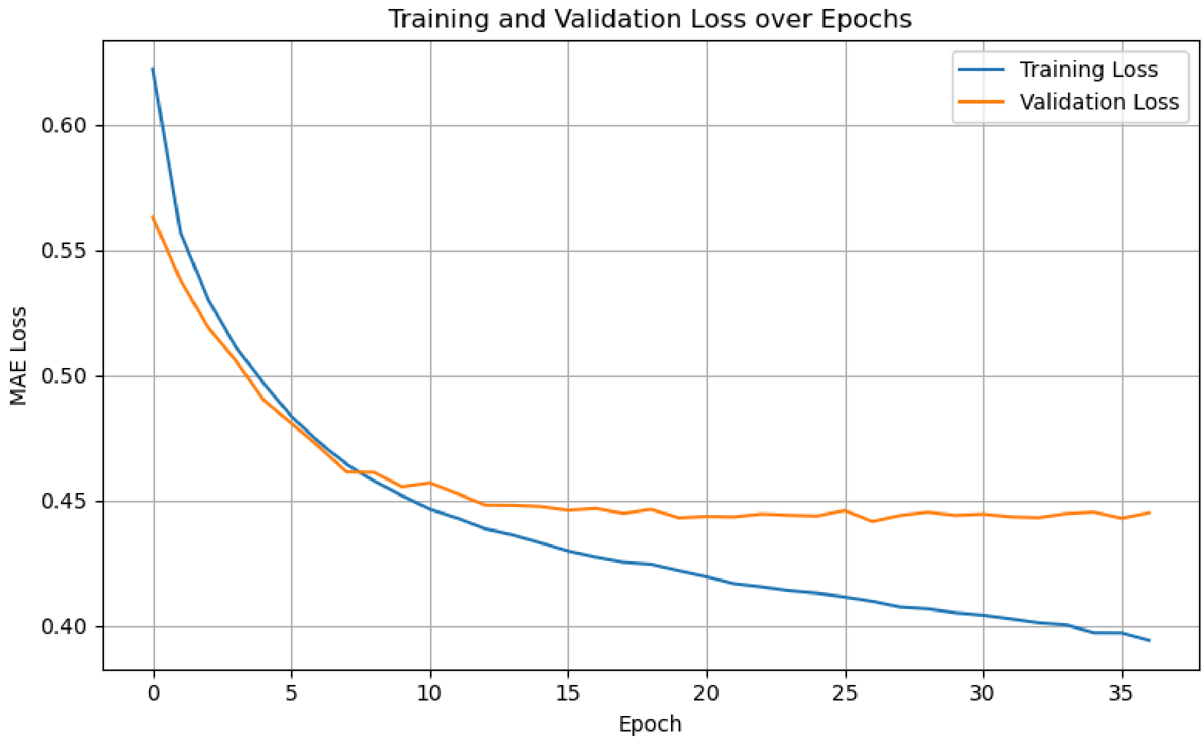

Figure 4: Loss Curve for Tuned Feed Forward Neural Network

Notes: The refined model was evaluated by comparing the minimum loss (MAE) for different architectures. The lowest loss was generated with two dense layers with [128, 64] neurons and a learning rate of 0.0005.

Table 9: Performance Metrics for Tuned Feed Forward Neural Network

Outcome Variable	R^2 Score	RMSE	MAE
Population Change %	0.362	0.940	0.672
Employment Change %	0.728	2.507	1.587
Establishment Change %	0.644	2.546	1.807
GDP Growth %	0.142	8.367	4.853

Figure 4 and Table 9 shows the performance of the neural network. Figure 4 indicates that validation loss plateaus after 15 epochs and remains close to training loss until early stopping ended training after 36 epochs. The R^2 of population change is 0.362 and GDP growth is 0.142, which is an improvement from the linear regression and Ridge regression models, further shown in Figure 6 below. The Ridge regression model has a higher R^2 score than the neural network for employment change and establishment change. The RMSE and MAE increased for these variables between the Ridge regression and neural network, but were lower for population change and GDP growth.

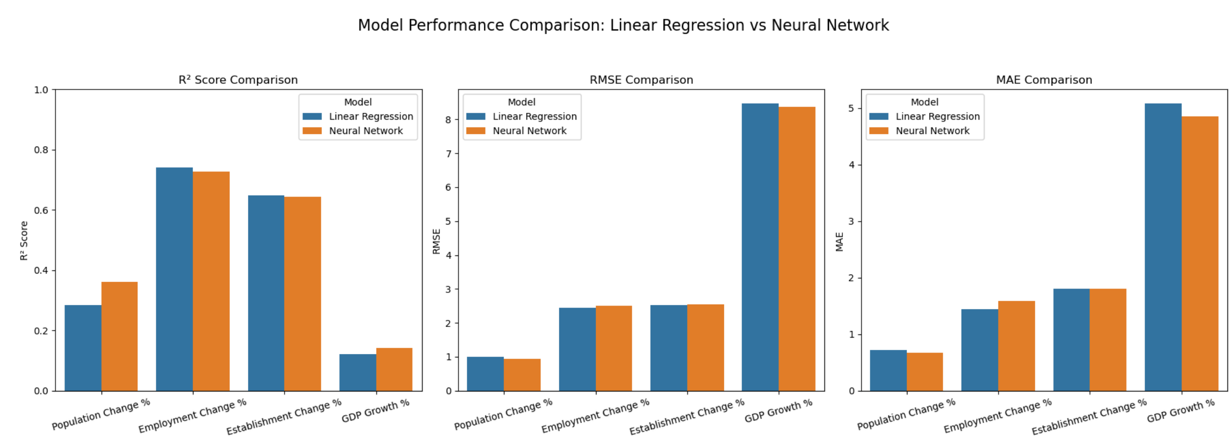

Figure 6: Comparing Performance Metrics between Ridge Regression and Neural Network

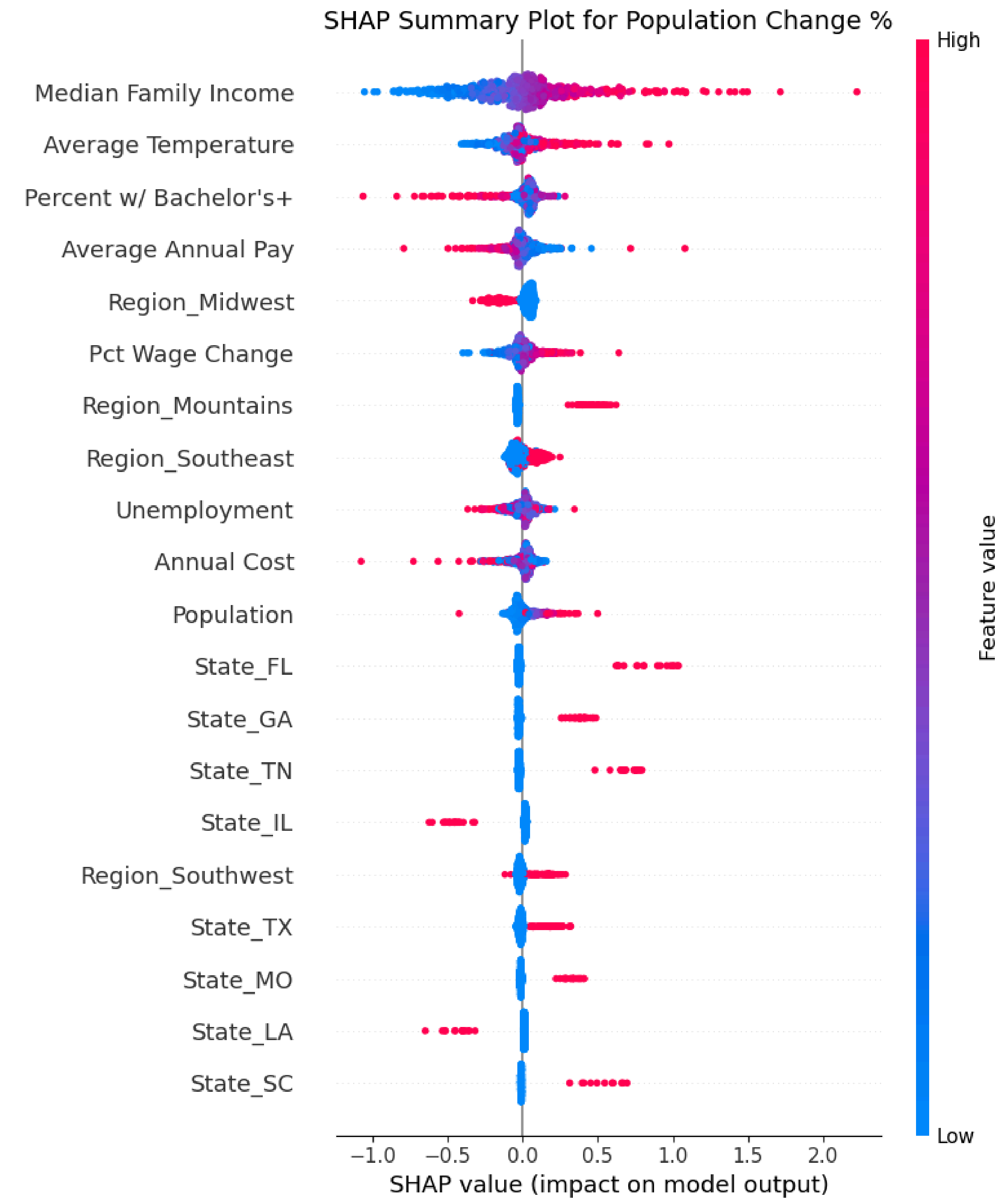

Figure 7: SHAP Summary Plot for Population Change

Figure 7 shows the neural network's top predictors for population growth using SHAP values. Compared to the regression models, the continuous variables appear higher in the order influencing the predicted values. Most notably, the median family income, average temperature, and percent of a county's population with a bachelor's degree or higher influence the model's prediction of population growth. Figure 8 below additionally shows the SHAP summary plot for GDP growth. Just like Figure 7, continuous variables influenced the predictions for GDP growth more than the categorical geographic features. Many of the top predictors seem to both increase and decrease the model's output.

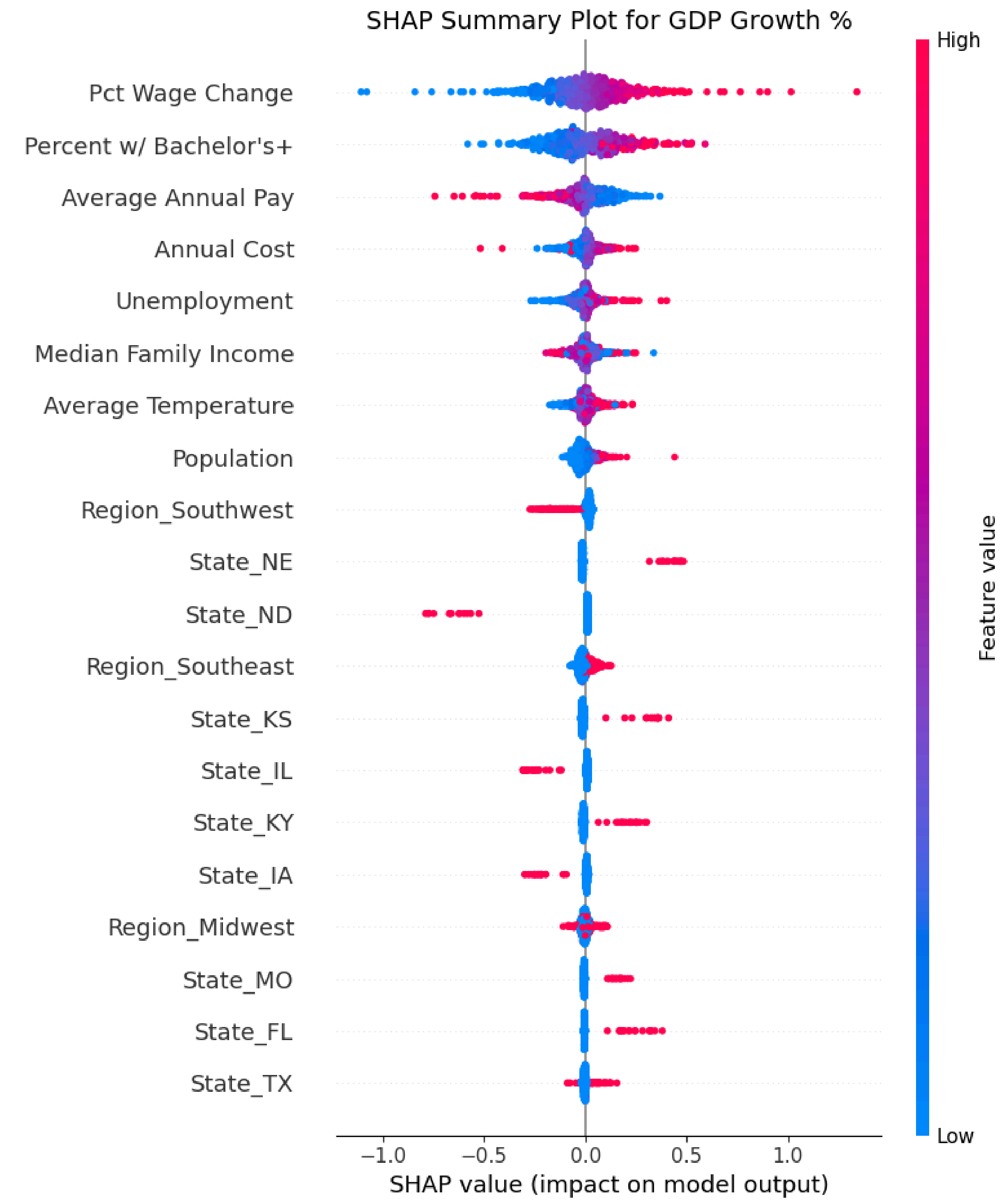

Figure 8: SHAP Summary Plots for GDP Growth

Discussion and Conclusions

This study aimed to identify the most important county-level geographic, economic, and demographic predictors of growth in U.S. counties, including changes in population, employment, establishment counts, and GDP. Three models were used to explore this relationship: linear regression establishing a baseline, Ridge regression improving this relationship using regularization, and a feed-forward neural network capturing nonlinear patterns hidden within the data.

Results showed that the Ridge Regression slightly outperformed linear regression across all outcome variables, offering more stable and generalizable predictions while retaining the interpretability of the model. The feed-forward neural network achieved slightly higher R^2 values for population growth and GDP growth, but suffered slightly in capturing the variation in employment growth and establishment growth.

These findings suggest that regularization is important in modeling the socioeconomic growth indicators with correlated predictor variables. The Ridge regression model improved performance by producing more stable coefficients and preventing overfitting. This ensures that the model was able to actually learn from the training data, and did not simply copy the patterns inherent to it.

The feed-forward neural network modestly improved performance in predicting the population growth and GDP growth in U.S. counties, but demonstrated marginally worse performance in predicting employment and establishment growth when compared with the Ridge regression model. This suggests that there is likely a nonlinear relationship obscured between the input features and population and economic growth in United States counties, which is reflective of the complex relationship between different socioeconomic factors in the real world.

Additionally, a linear model seems most appropriate for capturing the relationship between the input features and employment and establishment growth as the Ridge regression model explains more of the variance and maintains lower error rates for these growth predictors than the neural network.

The coefficients and SHAP values reveal which factors contribute most to predicting growth in United States counties. Different states and regions, particularly in the southeastern and mountain states on the West Coast, appear to have a significant effect on predictions for these different growth outcomes. This highlights a major limitation of this study. The use of one-hot encoded variables by geographic region shows which states and regions are growing, but it fails to capture the complexities of why these different places are growing. Different policies, tax rates, and legislation are just some examples of the factors common to counties in different states that are not explicitly captured in this dataset. Additionally, the overall weak predictive power of these models suggests that more features are needed to better model what factors actually drive growth in United States counties.

In conclusion, this study set out to identify the key economic, demographic, and geographic factors associated with county-level growth across the United States, using both interpretable and flexible machine learning models. These findings support the idea that county-level growth is influenced by both linear and complex nonlinear interactions among socioeconomic variables. However, the limited predictive power of all models highlights the need for more granular and policy-specific features to better understand the mechanisms of regional development and inform future planning and investment decisions.